

Networking & Cybersecurity Track
Simulated Employment Program

Lab Report:
The Cybersecurity Sorcery Cube Scatter Quizlet

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Objectives

Identify the three dimensions of the Cybersecurity Sorcery Cube and the elements of each dimension.

Required Resources

- PC or mobile device with internet access

Background / Scenario

The McCumber Cube was developed by John McCumber in 1992. It is a framework that is used to establish and evaluate information and information systems security. The framework relies on the cybersecurity professional to identify information assets with a focus on the core principles of cybersecurity: confidentiality, integrity, and availability. The model is based on three dimensions, each having three elements.

Dimension One: Cybersecurity Principles (CIA)

Dimension 1 - Cybersecurity Principles (CIA)	
Confidentiality	Assurance that sensitive information is not intentionally or accidentally disclosed to unauthorized individuals.
Integrity	Assurance that information is not intentionally or accidentally modified in such a way as to call into question its trustworthiness or reliability.
Availability	Ensuring that authorized individuals have both timely and reliable access to information and information systems.

Dimension Two: Information (Data) States

Dimension 2 - Information (Data) States	
Storage	Data at rest - stored in memory, on a drive, or USB flash drive.
Transmission	Transferring data between systems.
Processing	Performing operations on data such as modification, backup, and corrections.

Dimension Three: Security Countermeasures / Safeguards

Dimension 3 - Security Countermeasures / Safeguards	
Policy and Practices	Administrative controls such as information security policies, procedures, guidelines, and management directives.

Human Factors	Ensuring that users of information systems are aware of their roles and responsibilities. Requires awareness and education programs.
Technology	Software- and hardware-based solutions designed to protect information systems, such as antivirus, firewalls, and IDS/IPS systems.

Quizlet Review

Go to the Quizlet website and review the Cybersecurity Essentials - Cybersecurity Sorcery Cube. Although an account for quizlet.com is not required, you may want to sign up to avoid having to close the sign-up dialog box every time it pops up.

a.	Review the list of terms and definitions.
b.	Select the Flashcards icon. Test your knowledge by going through all ten flashcards.
c.	Click Back.
d.	Click Match under the Play heading. When prompted, click to start the game.
e.	Drag each term to its definition.

Reflection Questions

After completing the Quizlet activity, answer the following questions:

1.	What are the three core principles of cybersecurity as represented in Dimension One of the McCumber Cube?
Answer: The three core principles of cybersecurity in Dimension One are Confidentiality, Integrity, and Availability, commonly known as the CIA Triad. Confidentiality ensures that sensitive information is only accessible to authorized individuals. Integrity ensures that information is accurate and has not been tampered with. Availability ensures that authorized users can access information and systems when they need them.	
2.	List the three data states in Dimension Two and give a real-world example of each.
Answer: The three data states are: (1) Storage - data at rest, for example files saved on a hard drive or a USB flash drive. (2) Transmission - data moving between systems, for example an email being sent over the internet or a file being uploaded to the cloud. (3) Processing - data actively being used, for example a banking app calculating your account balance or a user editing a document on their computer.	
3.	Why is the Human Factors safeguard considered one of the most important countermeasures in cybersecurity?
Answer: Human Factors is considered one of the most critical safeguards because humans are often the weakest link in any security system. Even the most advanced technology can be bypassed if users fall for phishing emails, use weak passwords, or mishandle sensitive data. Security awareness training and education programs help ensure that people understand their roles and responsibilities, making them less likely to unintentionally compromise the security of a system.	

4. How does the McCumber Cube help cybersecurity professionals think about protecting information assets?

Answer: The McCumber Cube provides a structured three-dimensional framework that helps cybersecurity professionals evaluate information security comprehensively. By considering all three dimensions simultaneously - the security goals (CIA Triad), the data states (storage, transmission, processing), and the safeguards (policy, human factors, technology) - professionals can identify gaps in protection and design well-rounded security strategies that address every possible combination of risk, rather than focusing on only one aspect of security.